

# Vivek Jadhav

📞 7972064632 | ✉ [vivekjadhavofficial01@gmail.com](mailto:vivekjadhavofficial01@gmail.com) | [in linkedin.com/in/vivek-jadhav-17b408254/](https://www.linkedin.com/in/vivek-jadhav-17b408254/) | [github.com/vivekjadhav07ml](https://github.com/vivekjadhav07ml) | 

## SUMMARY

GenAI and Automation enthusiast with hands-on expertise in **LLMs**, **LangChain**, and **agentic frameworks**. Skilled in developing **AI/ML models**, rapid prototyping, and scalable APIs for real-world logistics optimization. Passionate about building intelligent, high-performance solutions that streamline global supply chains.

## EDUCATION

### Symbiosis Institute Of Technology Pune

*Bachelor of Technology Robotics & Automation*

2022-2026

CGPA : 8.3/10

## EXPERIENCE

### Research Intern

July 2024 – April 2025

*Symbiosis Centre for Applied Artificial Intelligence (SCAAI)*

*Pune, India*

- Developed a predictive maintenance solution for vertical milling machines using multi-modal machine learning, integrating sensor data to improve accuracy and machine lifespan
- Achieved 97% prediction accuracy and an F1-score of 96%, increasing tool lifespan by 20% through proactive maintenance.

### Social Volunteer

JAN 2024 – MAY 2024

*Being Volunteer Organization (NGO)*

*Location*

- Coordinated and managed social activities and community outreach events.
- Assisted in project management tasks, including planning and organization
- Activities : Wall-Beautification, Educational sessions for underprivileged children, Feed The Needy.

## PROJECTS

### Intelligent Facial Recognition for Smart Attendance System | *OpenCV, Raspberry Pi 5, dlib, Streamlit*

- Integrated Multi-Face Detection and Recognition using OpenCV frameworks to enable real-time, contactless attendance tracking, created custom dataset of 85 students
- Deployed a Streamlit app to visualize attendance records and analytics with clear graphs.
- Outperformed Vivo's facial recognition on identical twins through optimized embedding generation and detection.

### Hybrid Multi-Modal Neural Architecture for Enhanced Wind Turbine Power Forecasting | *Tensorflow*

- Developed a multi-modal deep learning framework combining CNN-LSTM for sensor data and a dense autoencoder for theoretical inputs to forecast wind turbine power output.
- Engineered cyclical wind direction encodings and lag features; achieved optimized results through hyperparameter tuning ( $R^2$ : 0.95, MAE: 180).

### Explainable AI: Scene Classification and Grad-CAM Visualization | *ResNet-18, Grad-CAM*

- Built a ResNet-18 model to classify scenes (e.g., forests, glaciers, streets) with 86% accuracy and strong generalization.
- Grad-CAM activation maps overlay key decision regions, delivering transparent, trustworthy AI for environmental monitoring, urban planning, and autonomous navigation.

### Nebula-X: AI Chatbot with Live Search (Gemini 1.5, Serper API, LangChain) | *Gen AI, LLMs, LangChain*

- Engineered Nebula-X by integrating Gemini 1.5 LLM with the Serper web-search API through LangChain pipelines, enabling dynamic, live search-driven conversational AI with sub-200 ms response times.
- Designed and implemented Model Context Protocols (MCPs) for orchestrating prompts and tool chains, reducing integration complexity by 40%.
- Achieved 92% user satisfaction in beta trials and published a detailed demo and write-up on LinkedIn.

## TECHNICAL SKILLS

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**Languages:** Python, C++, SQL

**Frameworks & Libraries:** LangChain, RAG, TensorFlow ,N8N

**Web & API:** Flask, FastAPI, Figma ,HTML,CSS

**Developer Tools & DevOps:** Git, GitHub, Docker, Kubernetes, VS Code, PyCharm, Visual Studio

**Databases:** PostgreSQL, MySQL

**Machine Learning & AI:** Deep Learning, Transfer Learning, NLP, Computer Vision,LLMs, Data Preprocessing

**Automation & Control:** PLC Programming, Industrial Protocols, Arduino, Cisco Packet Tracer

**Project Coordination:** Agile methodologies, Project Management, Problem Solving

## PUBLICATIONS

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- **Enhancing Tool Wear Classification Using SMOTE with Ensemble and Deep Learning Techniques**  
*MethodsX Journal* — Paper ID: MEX-D-25-00448 [Under Review]
- **Pseudo-Colorization of SAR Imagery Using GANs with VAE Generators for Enhanced Visual Interpretation in Remote Sensing**  
*16th International IEEE Conference on Computing and Networking Technologies* — Paper ID: 9834
- **An Approach for High-Speed Network Infrastructure Design for Health Services**  
*7th International Conference of Energy, Power and Environment*
- **Multi-Modal Neural Architecture for Enhanced Wind Turbine Power Forecasting**  
*1st International Conference on Power and Intelligent Control System*
- **Traffic Sign Detection and Recognition for Autonomous Vehicle Navigation by using Satellite Data and Deep Learning Techniques**

## CERTIFICATIONS

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**Decentralized Finance (DeFi) Infrastructure**

*Duke University*

*Skills: Blockchain, FinTech, Algorithms*

**Microsoft Azure SQL**

*Microsoft*

*Credential ID: 12GUHYDGYAN6*

**The Complete LangChain & LLMs Guide**

*Coursera*

*Credential ID: CDART3Y2XC3X, Skills: LangChain Integration*

**GenAI for Customer Communication**

*Coursera*

*Credential ID: Y28KW8KAGS9Z, Skills: Generative AI*

**TensorFlow 2.0 Deep Learning and AI**

*Udemy*

*Credential ID: 0004, Skills: TensorFlow, Deep Learning, AI Implementation*

## BLOGS

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**Retrieval-Augmented-Generation-framework-with-Langchain-for-chat-with-Docs**

*[www.Rag\\_hat.com/blog-post-link](http://www.Rag_hat.com/blog-post-link)*